

Firm Productivity, Panel Estimation, and Export Dynamics

An Applied Econometrics Portfolio Project in Stata

Yurou Liu

Applied-Econometrics-with-Stata

Methods: pooled OLS, firm fixed effects, two-way fixed effects, random effects, Hausman testing, two-step System GMM, TFP construction, and export-participation analysis.

Executive Summary

This project studies firm-level production and export behavior using an unbalanced panel of 2,199 plants observed between 1981 and 1991. The analysis begins with descriptive evidence on exporter productivity and capital intensity, then estimates a Cobb-Douglas production function using pooled OLS, fixed effects, random effects, and dynamic-panel System GMM. The final section constructs total factor productivity (TFP) and examines whether more productive firms are more likely to export.

The central empirical result is that time-invariant firm heterogeneity matters substantially. Fixed-effects estimates are smaller than pooled OLS estimates, and the Hausman test strongly rejects the random-effects orthogonality assumption. Although System GMM is designed to address dynamic endogeneity, the AR(2), Hansen, and Difference-in-Hansen diagnostics reject the validity of the reported specification. Accordingly, the fixed-effects model is treated as the most credible production-function estimate in this application.

Finding	Evidence	Interpretation
Exporters are larger and more productive	Mean log value added: 10.19 vs. 8.64	A raw exporter productivity premium of about 1.55 log points
Firm heterogeneity is important	Between-firm variation exceeds within-firm variation	Persistent technology and managerial differences motivate firm fixed effects
FE is preferred to RE	Hausman $\chi^2(2)=236.01$, $p<0.001$	Firm effects are correlated with capital and labor choices
System GMM diagnostics fail	AR(2), Hansen, and Difference-in-Hansen p-values <0.01	Dynamic-panel estimates should not be interpreted as causal
Exporting is persistent	Lagged exporter coefficient ≈ 0.78	Prior exporters are much more likely to continue exporting

1. Data and Panel Structure

The dataset contains 10,060 firm-year observations from 2,199 plants over 11 annual periods (1981-1991). The panel is unbalanced: firms are observed for between one and eleven years, and 230 plants (10.46%) appear in all eleven periods.

Panel feature	Value
Firms	2,199
Firm-year observations	10,060
Years	1981-1991
Maximum periods per firm	11
Median periods per firm	4
Balanced-panel firms	230 (10.46%)

2. Exporter Characteristics

An exporter indicator equals one when export sales are positive. Exporters account for 805 observations, compared with 9,255 non-exporter observations. Exporting firms have substantially higher average output, capital, and labor.

Variable	Non-exporters	Exporters	Difference
Log value added (lnv)	8.642	10.193	1.551
Log capital (lnk)	7.776	9.123	1.347
Log labor (lnl)	8.041	9.205	1.164
Log capital-labor ratio	-0.264	-0.082	0.182

The raw exporter productivity premium is approximately 1.55 log points and is statistically significant in a two-sample t-test. Exporters also exhibit a higher capital-labor ratio, suggesting more capital-intensive production technologies.

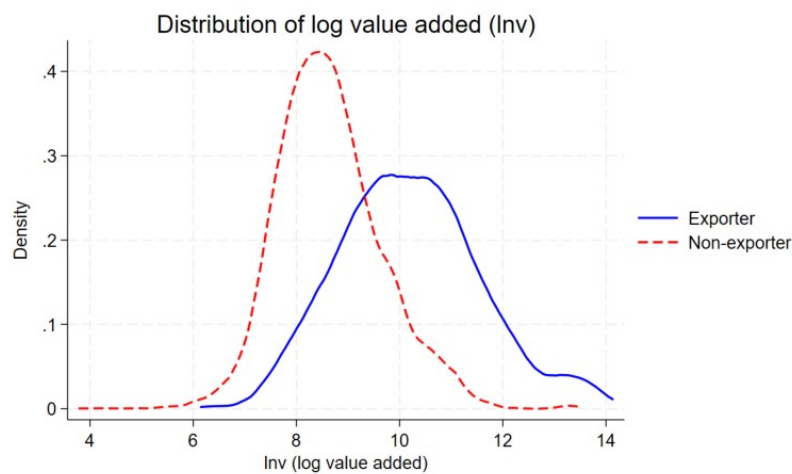


Figure 1. Distribution of log value added by export status.

3. Variance Decomposition

For output, capital, and labor, between-firm variation is considerably larger than within-firm variation. This pattern indicates that persistent firm-level differences dominate year-to-year changes and provides a strong empirical rationale for controlling for firm fixed effects.

Variable	Overall SD	Between SD	Within SD
Log value added	1.129	1.069	0.393
Log capital	1.172	1.131	0.385
Log labor	1.077	1.030	0.375

4. Production-Function Estimates

4.1 Pooled OLS

The baseline Cobb-Douglas specification regresses log value added on log capital and log labor using heteroskedasticity-robust standard errors. The estimated capital and labor elasticities are 0.231 and 0.744. Their sum is 0.975, close to constant returns to scale, although the formal test rejects an exact sum of one.

Pooled OLS is unlikely to be consistent because unobserved firm productivity is omitted and is likely correlated with input choices. More productive firms tend to employ more capital and labor, so pooled OLS may attribute productivity differences to observed inputs.

4.2 Exporter-Control Specification

Adding exporter status reduces the capital coefficient from 0.231 to 0.211 and the labor coefficient from 0.744 to 0.727. The exporter coefficient is 0.421 and highly significant. The modest decline in the input coefficients indicates that exporter status captures some observed productivity differences, but it does not solve the underlying omitted-productivity problem.

4.3 Fixed Effects

The firm fixed-effects estimator removes time-invariant plant heterogeneity. With firm-clustered standard errors, the estimated capital elasticity is 0.219 and the labor elasticity is 0.529. The resulting returns-to-scale estimate is 0.748, indicating decreasing returns to scale.

A joint test strongly rejects the hypothesis that all firm effects are zero ($F(2,195, 7,862)=6.40, p<0.001$). Adding year fixed effects yields very similar elasticities (0.218 for capital and 0.524 for labor), showing that common annual shocks do not materially change the production-function estimates.

4.4 Random Effects and Hausman Test

The random-effects model estimates capital and labor elasticities of 0.245 and 0.616, with returns to scale of 0.861. Approximately 61.7% of the composite-error variance is attributable to firm-specific heterogeneity. However, random effects requires the firm effect to be uncorrelated with capital and labor.

The Hausman test rejects this assumption ($\chi^2(2)=236.01, p<0.001$). Therefore, the random-effects estimator is inconsistent in this setting, and fixed effects is preferred.

Estimator	Capital	Labor	Returns to scale	Assessment
Pooled OLS	0.231	0.744	0.975	Likely upward biased by omitted firm productivity
Firm FE	0.219	0.529	0.748	Preferred static specification
Two-way FE	0.218	0.524	0.742	Robust to common year shocks
Random effects	0.245	0.616	0.861	Rejected by Hausman test

5. Dynamic Panel Estimation with System GMM

A dynamic production function includes lagged output to model persistence in firm performance. The model is estimated using two-step System GMM with collapsed instruments and robust small-sample inference.

Variable	Coefficient	Std. error
Lagged log output	-0.2734	0.0351
Log capital	0.3201	0.0455
Lagged log capital	0.0214	0.0203
Log labor	0.3776	0.0546
Lagged log labor	0.1409	0.0295
Constant	4.3267	0.5616

5.1 Specification Diagnostics

Diagnostic	Statistic	p-value	Implication
Arellano-Bond AR(1)	$z=-2.18$	0.029	Expected after first differencing
Arellano-Bond AR(2)	$z=-3.21$	0.001	Rejects required no-AR(2) condition
Hansen overidentification	$\chi^2(28)=274.91$	<0.001	Rejects joint instrument validity
Difference-in-Hansen	$\chi^2(3)=186.47$	<0.001	Rejects additional level moments

The significant AR(2) test and the rejection of both Hansen tests indicate that the instrument set is not credible. The negative lagged-output coefficient is also difficult to reconcile with a persistent AR(1) productivity process. Consequently, the System GMM estimates should be viewed as a diagnostic exercise rather than a preferred causal specification.

6. TFP and Export Participation

Firm-level TFP is constructed as the residual from the estimated production function. Export participation is then modeled as a function of TFP, lagged export status, and year fixed effects, with standard errors clustered by plant.

Variable	Approx. coefficient	Economic interpretation
Estimated TFP	0.03	A one-unit increase in TFP is associated with about a 3 percentage-point increase in export probability
Lagged exporter	0.78	Export participation is highly persistent over time

The positive TFP coefficient supports the self-selection view of exporting: more productive firms are more likely to enter foreign markets. The large lagged-exporter coefficient also indicates substantial state dependence, consistent with sunk entry costs and persistent export relationships.

7. Interpretation and Limitations

- The descriptive exporter premium is not automatically causal. It may reflect self-selection into exporting, learning-by-exporting, industry composition, or other omitted characteristics.
- Firm fixed effects remove time-invariant heterogeneity but do not eliminate simultaneity between current input choices and time-varying productivity shocks.
- The System GMM results fail key specification tests. A stronger analysis would reconsider lag depth, instrument classification, instrument count, or alternative production-function estimators.
- The linear probability model provides an interpretable export effect but can generate fitted probabilities outside the unit interval; conditional fixed-effects logit or complementary approaches may be useful robustness checks.

8. Reproducibility

The accompanying Stata do-file reproduces the workflow using relative project paths, input validation, model storage, diagnostic tests, automatic graph export, and saved derived variables. The repository is designed so that the analysis can be rerun from the module root after placing the original dataset in the data directory.

Recommended repository structure:

```
02_Firm_Productivity_Panel_Analysis/  
  README.md  
  code/02_firm_productivity_panel_analysis.do  
  data/README.md  
  output/Firm_Productivity_Panel_Analysis_Report.pdf
```

Conclusion

The analysis demonstrates why estimator choice is central to empirical production-function work. Exporters are larger, more capital intensive, and more productive in the raw data, but a substantial share of variation reflects persistent firm heterogeneity. Fixed effects materially reduce the estimated input elasticities and are preferred to random effects by the Hausman test. Dynamic-panel methods are conceptually attractive, yet their credibility depends on specification diagnostics; here, those diagnostics reject the proposed System GMM instrument set. The final TFP analysis is consistent with productive firms self-selecting into export markets and with strong persistence in export participation.